Real-Time S+C+L- Band 134-Tb/s DCI Bidi Transmission with All Channels at 1.2-Tb/s Enabled by SiPh Transceiver

Baoluo Yan[1,2 †], Dongchen Zhang[1,2 †], Zhenqian Yang[1,2], Hu Shi[1,2 *], Mingqing Zuo[3], Dawei Ge[3], Jun Wu[4], Zhaolong Liao[4], Xingzhou Xu[5], Taili Wang[5], Hong Liu[1,2], Huan Chen[1,2], Qiang Qiu[1,2], Yan Zhao[1,2], Wenbo Yu[1,2], Zhenhua Feng[1,2], Zhiyong Zhao[1,2], Kezhi Qiao[1,2], Yong Chen[1,2] , Ruichun Wang[4], and Qiang Wang[1,2]

[1]State Key Laboratory of Mobile Network and Mobile Multimedia Technology, Shenzhen 518055, China
[2]WDM System Department of Wireline Product R&D Institute, ZTE Corporation, Beijing 100029, China
[3]Department of Fundamental Network Technology, China Mobile Research Institute, Beijing 100053, China
[4]State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Yangtze Optical Fibre and Cable Joint Stock Limited Company, Wuhan, Hubei, 430073, China
[5] ZTE Photonics Technology Japan Co. Ltd., Chiba City, 261-0023, Japan
*[†]These authors contributed equally to this work. *shi.hu@zte.com.cn*

Abstract: We demonstrate real-time transmission of 134-Tb/s capacity with all channels operating at 1.276-Tb/s, supporting bidirectional transmission across 15.75-THz triple-band over 75-km G.654.E fiber for data centre interconnection. © 2026 The Author(s)

1. Introduction and Research Motivation

The shortage of existing fiber resources for over-the-top providers is becoming increasingly apparent in data center interconnection (DCI)[1]. Achieving higher channel rates, ultra-broadband, and bidirectional WDM transmission on a single fiber represents an effective solution to reduce in per-bit cost, power consumption, and fiber leasing fees. Over the past two year, 12-THz super C+L- band optical transport network (OTN) has been commercially deployed in China. With advancing S-band fiber amplifiers and wavelength selective switches (WSS) technologies, the S-band is now prime for expansion. This has resulted in a series of impressive achievements based on offline processing, exceeding 150 Tb/s capacities and >1 Tb/s/λ rates across the S, C, and L bands. However, implementing high-order modulation formats beyond 64 quadrature amplitude modulation (QAM) signals and forward error correction (FEC) with high complexity on digital signal processing (DSP) application-specific integrated circuits (ASIC) is challenging. Furthermore, undetectable burst errors in offline processing contribute to a notable performance discrepancy between offline and real-time systems.

In 2019, using the complex wavelength conversion technique based on four-wave mixing could temporarily address the low maturity of S- and L-band transceivers and components[2]. Until 2022, NTT demonstrated a groundbreaking S-band real-time transceiver based on an S-band laser diodes, lithium niobate modulator and DSP-ASIC, achieving a channel rate of 500 Gb/s and a single-fiber capacity of 112.8-Tb/s [3]. Last year, we tried to use silicon photonics (SiPh) transceiver with output power ~10 dBm tunable laser instrumentation (Santec TSL-570) in S-band and demonstrated real-time 128.7-Tb/s transmission with 500 Gb/s PCS-16QAM in S band[4]. However, the transmitter OSNR and receiver sensitivity were severely constrained, limiting further increases in bit rate/λ.

In this work, we presented a total capacity of 134-Tb/s with 105-λ×1.276-Tb/s 135-GBd PCS-64QAM covering triple-band with 15.75-THz bandwidth bidi-transmission over 75-km ITU-T G.654.E fiber, only adopting doped fiber amplifiers(DFA). To the best of our knowledge, this represents the highest real-time bit rate in the S-band and sets a new record for the highest single-fiber real-time capacity across the S, C, and L bands.

2. Basic Principles and System Design

As shown in Figs. 1(b), our SiPh transceiver integrates the DSP-ASIC based on 5-nm complementary metal oxide semiconductor technology, transimpedance amplifier (TIA), driver (DRV), and SiPh modulator and receiver photonic integrated circuit (PIC) using optoelectronic multiple-chip module (OE-MCM) co-packaging technique, offering enhanced bandwidth. This assembly supports up to near 70-GHz O/E bandwidth, with a modem linerate of 1.276-Tb/s. Leveraging the wavelength-insensitive of SiPh modulators, optical signal modulation across S-band, C-band, and L-band is enabled. We developed a nano-packaged S-band ITLA with a 6.4-THz tuning bandwidth and output power >17.5 dBm. This is crucial for achieving 1.276-Tb/s and significantly improves the S-band transceiver output OSNR and receiver sensitivity. The S- band signal spectrum after IQ modulator locking and corresponding constellation diagram in BtB case are shown in Fig. 1(c). We also equipped a nano-packaged L-band ITLA with output power >18.5 dBm to compensate for the low long-wavelength responsivity of SiPh receivers.

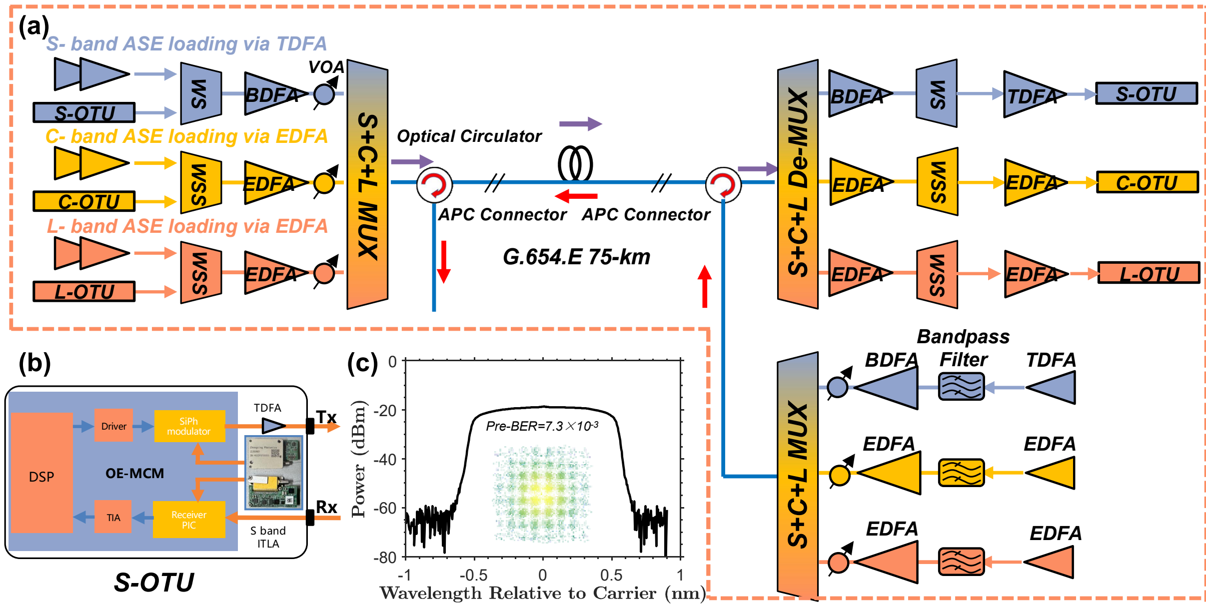

Fig. 1 (a) Setup of real-time bidi- 134 Tb/s transmission over the 75-km G.654.E link; (b) Schematic of our proposed S-band SiPh transceiver integrated with S-band nano-ITLA; (c) Spectrum of the 1.2 Tb/s signal in the S-band at the Tx side and corresponding constellation diagram in BtB case. (BDFA/TDFA/EDFA: Bi-/Tm-/Er- doped fiber amplifiers; APC:angled physical contact; OTU: optical transmitter unit)

We found applying OEQ[5], a bandwidth enhancement technique, at the transmitter WSS significantly improves 1.2 Tb/s transmission performance. Fig. 2(a) illustrates the process of determining the OEQ depth at 1492.80 nm for example, then we selected and applied the optimal parameters to all channels. The experimental setup for bidi- S+C+L-band transmission is shown in Fig. 1(a). The measured attenuation coefficient and the A_{eff} of G.654.E fiber are 0.159 dB/km and 125 μm^2 at 1550 nm, respectively. The each band cascaded optical amplifiers, shaped by corresponding WSS or waveshaper (WS), emulates 105 channels with a bandwidth 137.5 GHz /λ for the S-band (3.75-THz), C-band (6-THz), and L-band (6-THz). Limited by the BDFA/TDFA output power and increasing SRS, the S-band 6.4-THz transceiver spectrum was not fully utilised in the transmission experiment. The main interferences in such a system are Rayleigh backscattering (RB), Fresnel reflections (FR) and stimulated Raman scattering (SRS). We added different power pre-emphasis across inter-band and DFAs intra-band gain tilt to overcome SRS and RB effects on SNR flatness, then the average launch powers per channel for S-, C- and L-band are 6.23 dBm, 1.00 dBm, 1.82 dBm, respectively, as shown in Fig.2(b). These results originate from calculations by the power management algorithm, as detailed in our previous work [6]. The distinction of bidirectional S+C+L-band transmission in a single fiber lies in the simultaneous presence of forward and backward Raman amplification/absorption effects. Thus, effective control is required to ensure the input power profiles in forward and backward directions remain as consistent as possible, achieving equivalent signal transmission performance in both directions. As shown in Fig.2(c), it is evident that 15.75-THz transmission causes up to maximum 5-dB additional loss/gain in the S- and L-bands, respectively, making high launch power unnecessary for the L-band. Limited by the available number of S-band WS, we employed flat S+C+L-band ASE loading for the backward direction and ensured that the total power and frequency range in each band matched those of the forward direction to achieve an equivalent simulation of bidi- transmission. Notably, the S-band seed light should first pass through a band-pass filter to limit the amplification range, since the primary gain region of BDFA and TDFA are located at the shorter wavelength range (Fig. 2(d)). The BDFA and TDFA gain and NF spectra were acquired at a WDM total input power of 0 dBm. BDFA offers up to 21 dBm output power with lower noise level than TDFA's 19 dBm, so BDFA is recommended as the booster amplifier. We also employed APC connectors in the link to mitigate connector-induced FR. Additionally, an extra optical amplifier was placed before the SiPh receiver to ensure signal power > 0 dBm and avoid received power penalties.

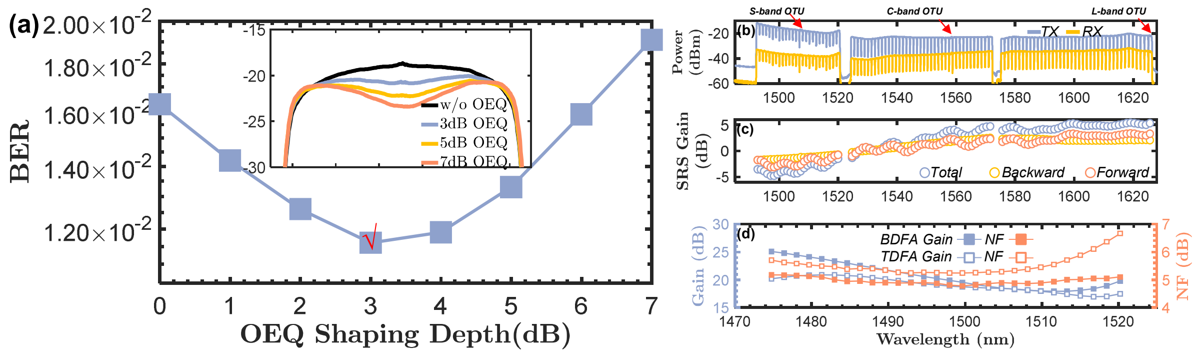

Fig. 2 (a) Pre-FEC BER after fiber transmission versus OEQ shaping depth at 1492.80 nm; The inset presents the transmitted spectra for various OEQ strength settings. (b) Overall spectra at the transmitter and receiver sides with a wavelength resolution of 0.03nm; (c) Forward and backward SRS gain; (d) Gain and noise figure (NF) versus λ for BDFA and TDFA under total input power 0dBm with WDM configuration.

3. Results and Discussion

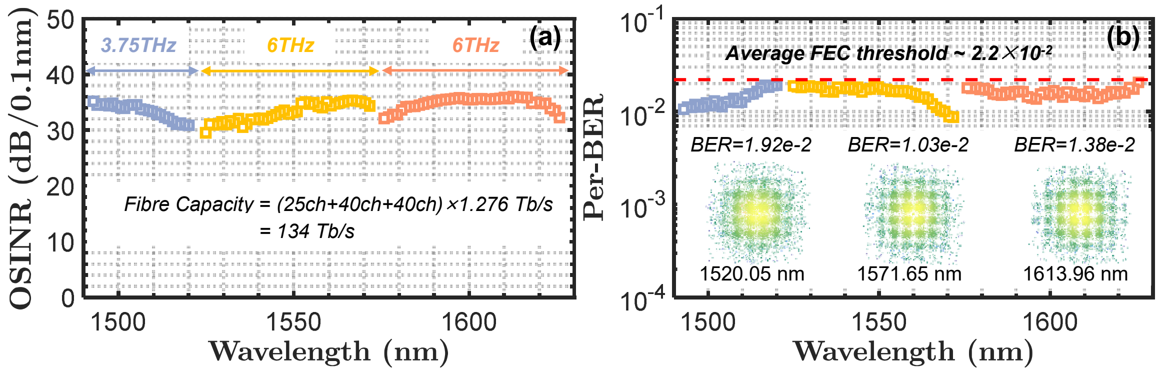

Fig. 3 (a) Measured signal-to-interference-plus-noise ratio (OSINR) and (b) pre-FEC BER values of all 105 channels in forward direction after 75-km G.65E fiber transmission. Inset is the recovered PCS-64QAM constellation diagram

We measured all 105 channels by tuning the transceiver wavelength to replace each ASE loading channel, carefully adjusting the launch power to ensure it was consistent with the original ASE channel power. Note that, the current setup represents a conservative emulation compared to the full modulated loader due to nonlinear penalty difference[7]. Some claimed capacity records with similar setups are referenced in [8,9]. Considering the effects of launch power, NF, forward and backward SRS, and contributions from RB and FR, a relatively flat OSINR was achieved at the receiver, averaging 33.83 dB (Fig.3(a)). The measured per-BER values for each channel were all below the oFEC threshold, i.e., 2.2×10^{-2}.

4. Conclusion

Based on the design of wavelength-insensitivity 70-GHz SiPh PICs, high power S, C, L-band ITLAs, real-time 5-nm DSP-ASIC, OEQ techniques and bidirectional transmission with power pre-equalization, we demonstrated the highest real-time data rate of 1.276-Tb/s in the S-band and achieved the 134-Tb/s real-time G.654.E fiber transmission across S+C+L-band 15.75-THz-wide, with all the channels operating at 1.276-Tb/s PCS-64QAM. This demonstrates the S-band transmission technology's transition from offline verification to real-time commercial demonstration, with potential pioneering applications in medium-to-short-haul high-capacity scenarios such as inter-data-center links or metro networks.

References

[1] C. Zhu, *et al.*, OFC 2023, W3H.5.
[2] T. Kato, *et al.*, ECOC 2019, PD1.7.
[3] F. Hamaoka, *et al.*, OECC 2022, PDP-A-3.
[4] A. Zhang, *et al.*, Opt. Express. 31(16), 34041(2025).
[5] M. Nakamura, *et al.*, SPPCom. 2023, SpW4E.4.
[6] B. Yan, *et al.*, CLEO 2025, SS134_3.
[7] S. Searcy, *et al.*, OFC 2022, W1G.6.
[8] D. Ge, *et al.*, OFC 2024, M3J.2.
[9] Y.Frignac, et al., ECOC 2023, M.A.5.2.